AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

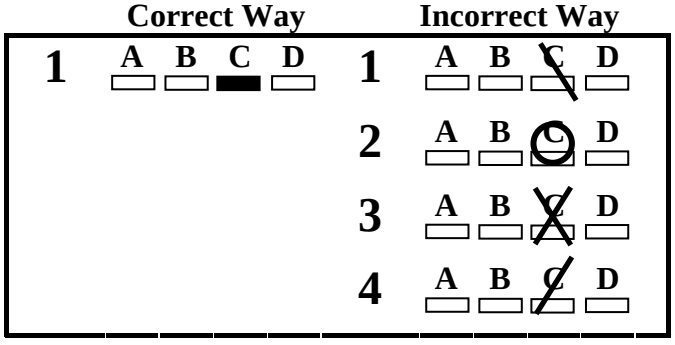
CLASS IX EXAMINATION 2010

Mathematics Paper I

Time allowed: 40 minutes Marks 30

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.
6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.
7. You may use a simple calculator if you wish.

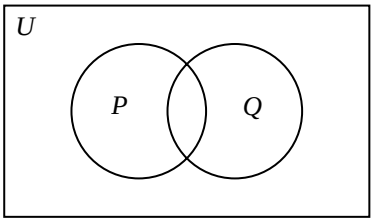
1. If $P = \{0, 2\}$ and $Q = \{0, 1\}$, then one of the ordered pairs of $P \times Q$ is

   A. (0,2)
   B. (1,0)
   C. (1,2)
   D. (0,1)

2. For two sets $A = \{0,1\}$ and $B = \{1,2\}$, which one of the following is a function from A to B ?

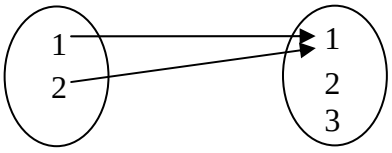
   A. $\{(1,1)\}$
   B. $\{(0,2),(1,2)\}$
   C. $\{(0,1),(0,2),(1,2)\}$
   D. $\{(0,1),(0,2),(1,1),(1,2)\}$

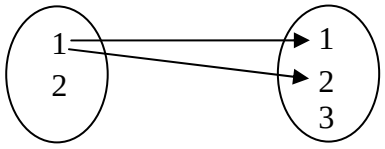
3. Consider the following Venn diagram. $P \cup (Q \cap P)$ is equal to

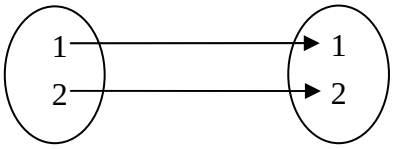


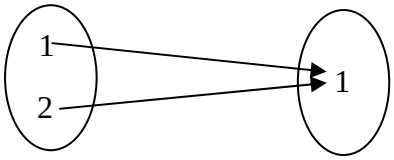
   A. P
   B. Q
   C. $P - Q$
   D. $Q - P$

4. Which one of the following figures represents an injective function?

   A.


   B.


   C.


   D.


5. If $i = \sqrt{-1}$, then i^3 is equal to

   A. –1
   B. 1
   C. $-i$
   D. i

6. In decimal fractions, $\sqrt{8}$ is a fraction which is

   A. terminating and recurring
   B. non-terminating and recurring
   C. terminating and non-recurring
   D. non-terminating and non-recurring

7. $\left(\frac{a^{-1}}{a}\right)^2$ is equal to

   A. 1
   B. a^4
   C. $\frac{1}{a^2}$
   D. $\frac{1}{a^4}$

8. The conjugate of $-2-3i$ is equal to

   A. $2+3i$
   B. $-2+3i$
   C. $2-3i$
   D. $-2-3i$

9. If $\log 8 = x$, then the value of $\log 2$ is

   A. x^3
   B. 3
   C. $3x$
   D. $\frac{x}{3}$

10. Which one of the following is a polynomial?

    A. $\sqrt{2}y^2 - \frac{1}{2}$
    B. $y - \frac{1}{y}$
    C. $\sqrt{y} + 2$
    D. $\sqrt{y^2 - 2}$

11. $(a-b+c)^2$ is equal to

    A. $a^2 + b^2 + c^2$
    B. $a^2 - b^2 + c^2$
    C. $a^2 + b^2 + c^2 + 2ab + 2bc - 2ca$
    D. $a^2 + b^2 + c^2 - 2ab - 2bc + 2ca$

12. If $P = \frac{1}{\sqrt{3}-\sqrt{2}}$, then $\frac{1}{P}$ is equal to

A. $\sqrt{3}+\sqrt{2}$
B. $-\sqrt{3}-\sqrt{2}$
C. $\sqrt{2}-\sqrt{3}$
D. $\sqrt{3}-\sqrt{2}$

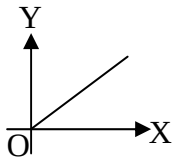
13. If remainder of $\left(x^2+2x-1\right)\div\left(x-1\right)$ is 2, then which one of the following is exactly divisible by $x-1$?

A. x^2+2x+2
B. x^2+2x-1
C. x^2+2x+1
D. x^2+2x-3

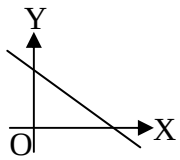
14. On factorization of $c^2-\left(d+2\right)^2$, we get

A. $\left(c+d+2\right)\left(c-d-2\right)$
B. $\left(c+d+2\right)\left(c-d+2\right)$
C. $\left(c-d+2\right)\left(c-d-2\right)$
D. $\left(c+d-2\right)\left(c-d-2\right)$

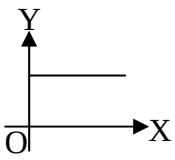
15. Which of the following graph shows a direct relation between x and y ?



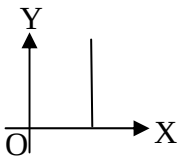
A.



B.



C.



D.

16. Which one of the following are inversely proportional to each other?

A. Side and area of a square
B. Age and weight of student
C. Radius and circumference of a circle
D. Number of days and number of workers for a task

17. For $x : y :: y : z$, z is known as

A. second proportional
B. third proportional
C. fourth proportional
D. mean proportional

18. If $\begin{bmatrix} 4 & -4 \\ 8 & 12 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix} + 3\begin{bmatrix} a & d \\ c & d \end{bmatrix}$, then the value of c is

A. -8
B. $-\frac{8}{3}$
C. $\frac{8}{3}$
D. 8

19. If $P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $Q = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $X = P \times Q$, then X is equal to

A. P
B. Q
C. QP
D. $Q^{-1}P$

20. For any three matrices A, B and C, $A \times (B \times C)$ is equal to

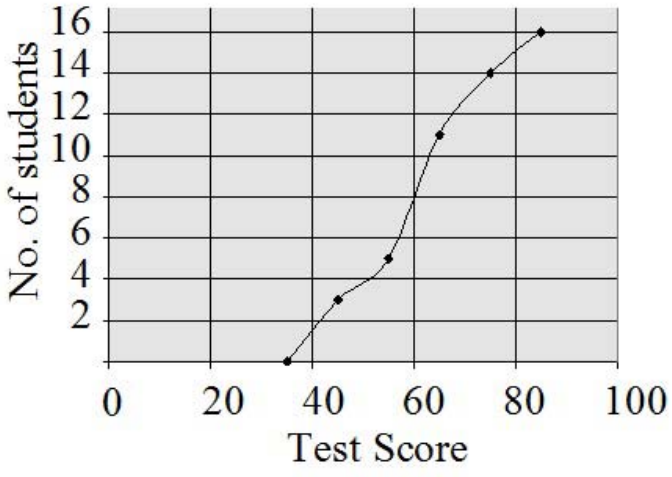
A. $(A \times B) \times C$
B. $(B \times C) \times A$
C. $A \times (C \times B)$
D. $(B \times A) \times C$

21. The median of four numbers is 6.5. If three of the numbers are 3, 5 and 9, then the 4th number is

A. 3
B. 4
C. 8
D. 10

22. In the following cumulative frequency curve, the test scores of 16 students are shown. The median test score is

A. 8
B. 50
C. 60
D. 65



23. The harmonic mean for 2, 3, 5, 9 is

A. $\dfrac{4}{2+3+5+9}$

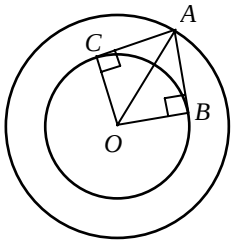
B. $\dfrac{1}{2}+\dfrac{1}{3}+\dfrac{1}{5}+\dfrac{1}{9}$

C. $\sqrt{\dfrac{2+3+5+9}{4}}$

D. $\dfrac{4}{\dfrac{1}{2}+\dfrac{1}{3}+\dfrac{1}{5}+\dfrac{1}{9}}$

24. The centre of both the circles given below is *O*. If *AB* and *AC* are tangents to the inner circle and $m\angle AOB = 20°$, then $m\angle OAC$ is equal to

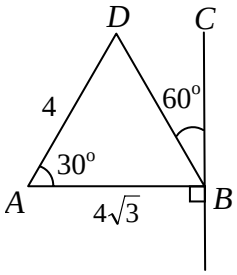
A. 20°
B. 70°
C. 90°
D. 110°



NOT TO SCALE

25. In the following figure $\overline{AB}$ is perpendicular to $\overline{BC}$. The length of $\overline{BD}$ is

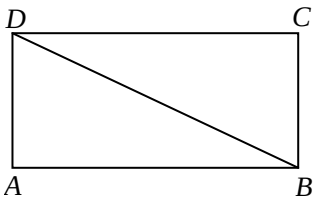
A. 2
B. $2\sqrt{3}$
C. 4
D. $4\sqrt{3}$



NOT TO SCALE

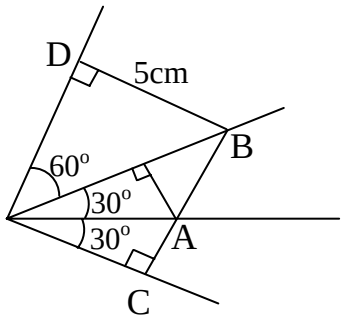
26. In the following figure ΔABD is congruent to ΔCDB. The corresponding angle of $\angle ADB$ is

A. $\angle ADB$
B. $\angle ABC$
C. $\angle CBD$
D. $\angle BDC$



27. In the following figure if AC = 2cm and BD = 5cm, then the length of AB is

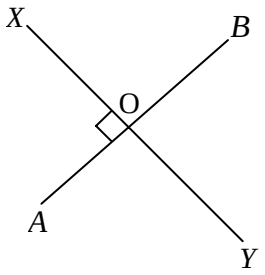
A. 2cm
B. 3cm
C. 4cm
D. 5cm



NOT TO SCALE

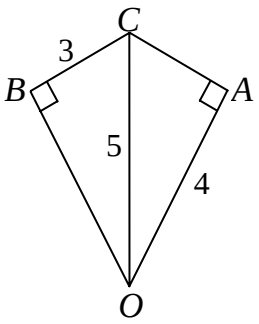
28. $\overline{XY}$ is the right bisector of $\overline{AB}$. If $m\overline{AB} = a$, then $m\overline{OA}$ is

A. $\frac{a}{2}$
B. a
C. $2a$
D. a^2



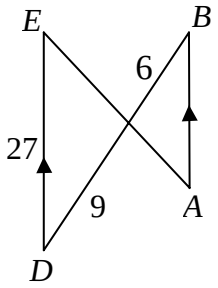
29. In the given figure if $\overline{OC}$ is the angle bisector of $\angle AOB$, then $m\overline{AC} + m\overline{BC}$ is equal to

A. 6
B. 7
C. 8
D. 9



30. In the following figure if $\overline{AB}$ is parallel to $\overline{DE}$, then the length of $\overline{AB}$ is

A. 6
B. 9
C. 15
D. 18



END OF PAPER

Please use this page for rough work